SPOTIFY LISTENING BEHAVIOR DASHBOARD

by Nathaniel Victor Ungco I December 31, 2025

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Dashboard Link:
https://upsystem-my.sharepoint.com/:u:/g/personal/ncungco_outlook_up_edu_ph/IQAfc6JMsgSgSobcNDeMfV-yAYfzkhSEjQ7sIp0Yab2LoKQ?e=cvsq8B

Github: https://github.com/nathanielv-ungco/spotify-listening-dashboard

BACKGROUND

I wanted to end the year 2025 with a project that feels both personal and practical. Every December, Spotify Wrapped comes out and it's one of the easiest examples to point to when people ask what data visualization is for. It takes something routine, like listening to music, and turns it into a story using design, summaries, and patterns.

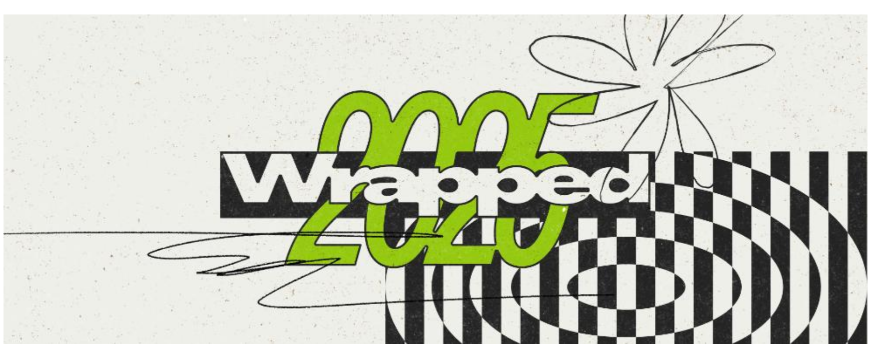

Figure 1: Spotify Wrapped 2025 Banner

That was the main push behind this project. If I could request my own Spotify data, could I build something similar from scratch?

I also recently learned data visualization in class, and I realized that making charts is only a small part of the work. In class, the dataset is usually prepared or clean enough. In real projects, the harder part happens before the visuals. You have to understand what the data actually represents, decide which metrics make sense, clean and transform fields, and make sure the report still behaves correctly when filters and interactions are used.

Spotify data felt like a good middle ground. The topic is familiar, and the dataset is personal, so I stayed interested while working on it. The coverage is long enough to show real changes too, since it spans 2021 to 2025. It also pushed me to work with JSON files and deal with real-world issues like time parsing and inconsistent text fields.

I treated this as my own Wrapped-style dashboard. It is not a copy of Spotify Wrapped's visuals. It is more like practice for summarizing, prioritizing information, and designing a report that tells a clear story.

I also used it to improve my confidence with Power BI. Before this, I understood features one by one, like Power Query, visuals, and measures. Building a full report made me connect everything. Layout, navigation, tooltips, interactivity, and short takeaways all had to work together.

This project is also meant to be honest about being a learning output. It is not a perfect analytics system. It documents what I can do right now, the decisions I made, and what I would improve next time, especially if I apply the same process to a more serious dataset, such as in business (e.g. HR, Operations, Finance), public sector, or research.

PROJECT PURPOSE AND SCOPE

WHY I MADE THIS?

This dashboard summarizes my Spotify listening activity from 2021 to 2025 using Power BI. My goal is simple. I wanted to turn [my] raw listening records into a report that is easy to explore and easy to explain.

I also treated it as practice for an end-to-end workflow. Import the data, clean it, create measures, design visuals, and write takeaways that match what the charts actually show.

WHAT THIS DASHBOARD ANSWERS?

- How has my listening changed over time?
- Who are my top artists, albums, and songs overall?
- When do I usually listen, based on day and hour?
- How long do I listen on a typical day?

DATA SOURCE AND HOW I DEFINE THE METRICS

The dataset came from my Spotify account data export (JSON). Each row represents a playback record in the export. More information about downloading your data Is here: Account privacy - Spotify

I mainly used two metrics:

- Minutes played
- Streams/plays

These are based on what the export provides. They may not match Spotify's official definitions used for Spotify Wrapped.

TOOLS USED

Power BI Desktop

- Power Query for cleaning and derived columns
- DAX for measures and insight text

WORKFLOW SUMMARY

I followed a straightforward process from raw files to a finished dashboard.

1. Imported Spotify JSON files into Power BI.
2. Combined and flattened the data into a usable table.
3. Cleaned types and created derived fields like Month-Year, day of week, and hour.
4. Built measures for totals, shares, and "top" items.
5. Designed the report pages and interactions.
6. Added tooltips and short takeaways.
7. Checked sorting, filtering behavior, and consistency across years.

DASHBOARD PAGES AND HOW TO USE THEM

OVERVIEW PAGE

The Overview page is for the big picture. It shows totals, the main time trend, and the top artists, albums, and song tracks. I kept the filtering simple using a Year slicer, then relied on clicking visuals to explore deeper, and also provided a slicer that will give the users the option to choose what metrics they wanted to analyze.

Typical interactions:

- Use the Year slicer to filter the report.

- Click an artist and/or album to update the album and/or song breakdown.
- Click points on the trend chart to focus on a specific month.

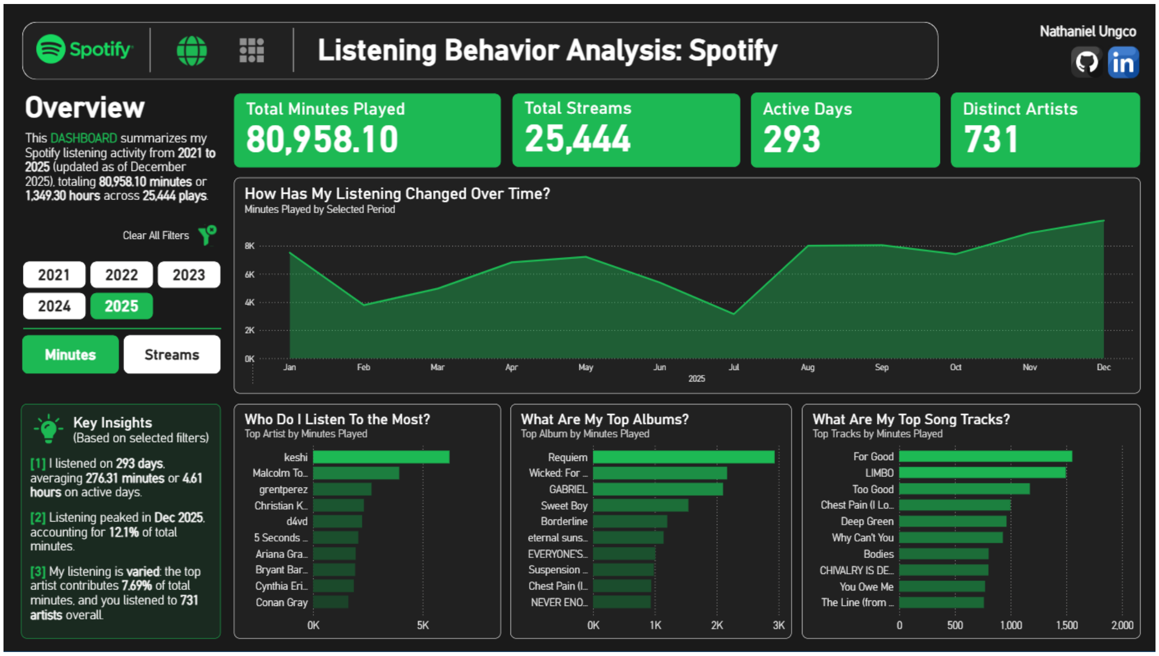

Figure 2: Overview Page

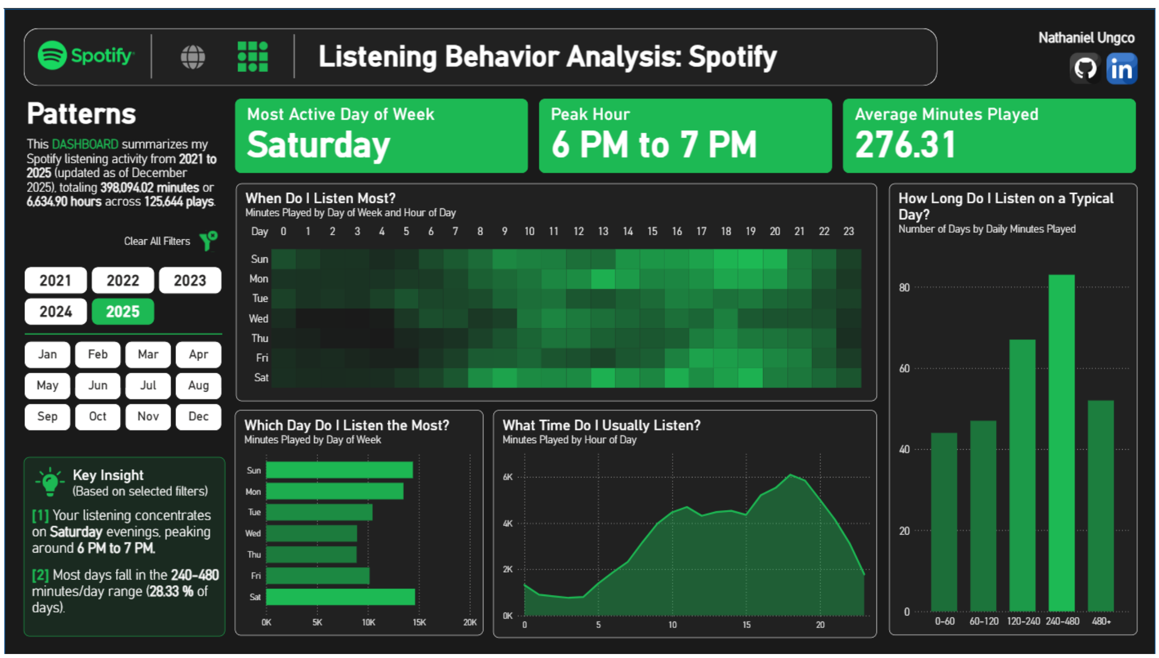

Figure 3: Patterns Page

PATTERNS PAGE

The Patterns page focuses on the Spotify listening routine. It highlights the most active days and hours and shows how listening time is distributed across days.

What to look for:

- Which day of the week is highest

- Which hour range is most active
- Whether most days are light listening days or there are many heavy listening days (distribution of minutes played buckets)

CUSTOMIZED TOOLTIPS

All charts in the dashboard (both pages) use customized tooltips to provide additional context without cluttering the visuals. These tooltips update dynamically based on the current filters and selections. To view the tooltips, you have to hover them on the charts.

The following are some of the previews of the tooltips.

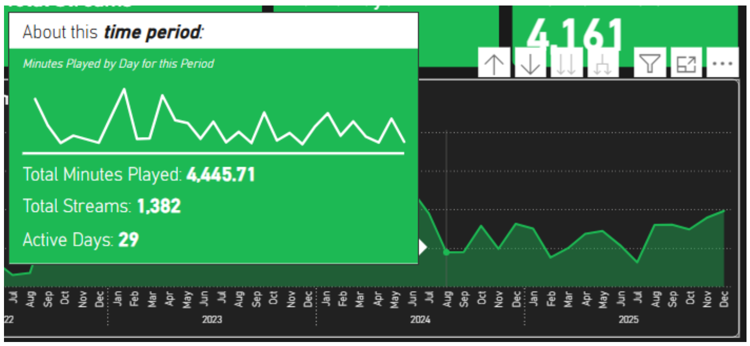

Figure 4: Tooltip for the trend chart in Overview Page

The tooltip (Figure 4) summarizes total minutes, total streams, active days, and daily listening behavior for the selected month or period.

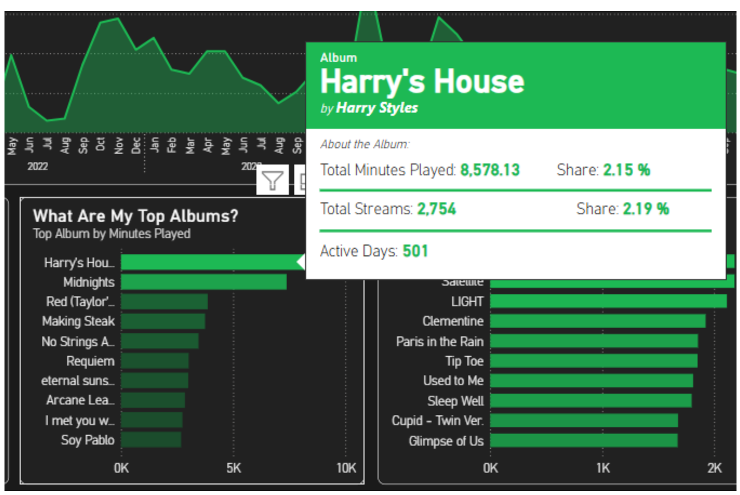

Figure 5: Tooltip for the bar chart of top albums in Overview Page

The tooltip (Figure 5) shows the total minutes, streams, listening share, and active days for the selected album.

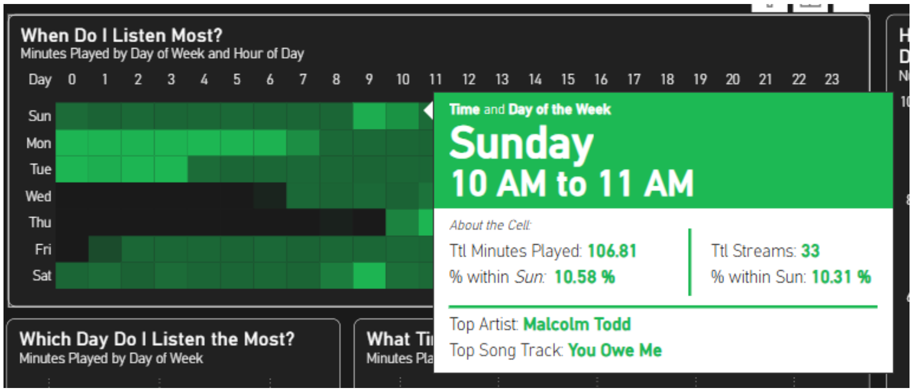

Figure 6: Tooltip for the heatmap chart in Patterns Page

The tooltip (Figure 6) shows the total minutes and streams for the selected time window, the share within the day, and the top artist and song for that hour.

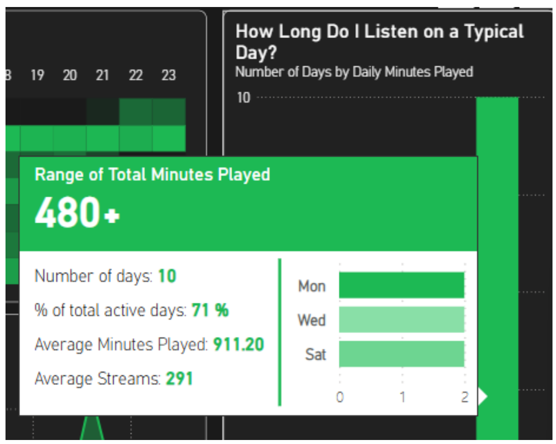

Figure 7: Tooltip for the histogram chart of daily minutes bucket in Patterns Page

The tooltip (Figure 7) shows the total minutes and streams for the selected time window, the share within the day, and the top artist and song for that hour.

CHALLENGES AND KEY DECISIONS

CLEANING AND FEATURE CREATION

The raw JSON needed flattening and consistent data types. Time fields also required extra care. I had to extract usable features like ListenDate, Month-Year labels, day of week, hour, and readable hour ranges.

Sorting was also a real issue at first. Month names do not automatically sort in calendar order. I needed a Month-Year sort key so the trend chart would not appear out of order.

MEASURES, FILTER BEHAVIOR, AND PERCENTAGES

Some measures were easy at first, like total minutes or total plays. The harder part was making sure they stayed correct under filtering and selection.

This became obvious in percentage measures, especially for distribution charts like daily minutes buckets. Early versions kept returning 100% because the denominator was not removing filters correctly. Fixing this taught me how important filter context is in DAX.

DATA MODEL CHOICE

For this project, I did not fully normalize the data into a complete star or snowflake schema. I mostly worked with a main listening history table, plus a small derived daily summary table for the bucket chart.

This was a conscious choice to keep the build simpler while learning. The dashboard works, but the structure has trade-offs:

- Some calculations are harder than they would be with a clean fact table and dimension tables.
- Text fields can create ambiguity, like repeated track names across different artists.
- Date handling and grouping required more manual work since I did not rely on a dedicated Date table.

LAYOUT AND CONTENT LIMITS

Space was a constraint. I could compute many metrics, but I could not show everything without making the report crowded.

So I made a few decisions:

- Keep the KPI set small.
- Use interactions and tooltips instead of adding more visuals.
- Separate the story into two pages so the Patterns page does not repeat the Overview.

LIMITATIONS

This project uses only the information available in Spotify's personal data export. It does not include all the metadata used in Spotify Wrapped. Some details that could better explain listening preferences, such as full genre information, playlist context, or consistent album identifiers, are missing or incomplete. Because of this, the analysis is limited to what can be derived from playback records alone.

The metrics used in the dashboard are simplified. Streams are counted as playback records, and minutes played are computed from recorded playback time. These measures are useful for exploring patterns, but they should not be treated as official Spotify metrics or compared directly with Spotify Wrapped results.

The dashboard focuses on describing listening behavior rather than explaining why it happened. The visuals can show when and how often I listen, but without additional context, they cannot fully explain the reasons behind those patterns.

Finally, the data itself is personal and recreational. While it works well for practicing data cleaning, modeling, and visualization, it does not directly address real-world decision-making problems. The main value of this project is in the workflow and skills developed, not in producing actionable insights.

REFLECTIONS & NEXT STEPS

WHAT I LEARNED I WHAT I WILL IMPROVE I WHAT FEATURES OR SKILL CAN I ADD/APPLY IN THE FUTURE

Working on this project made it clear to me that dashboards are not just about charts and visuals. Most of the work happens before anything appears on the screen. Cleaning messy data, defining metrics carefully, and making sure filters and interactions behave as expected matter just as much as the final design.

While I learned many new things throughout this project, it also made me more aware of what I still do not know and what I need to improve. Building the dashboard helped me understand where my workflow can be stronger and more intentional, especially when working with larger or more complex datasets.

If I apply this workflow again, these are the improvements and skills I want to focus on, both for this project and for future dashboard or data visualization work:

- Build a clearer data model earlier, using a star or snowflake schema depending on the problem.
- Standardize naming conventions for columns and measures from the start to avoid confusion later.
- Add stronger refresh and validation checks, such as row count comparisons and date range checks after updates.
- Plan key insights and page structure earlier so the final layout is more deliberate and story-driven, not just visually clean.

To conclude, this project was not just about analyzing my Spotify data, but about learning how to think more carefully about data preparation, modeling, and storytelling. These are areas I want to continue improving as I work on more dashboards and data projects.